Completing the transitive-tail alternative

Human Compression project: bounded L044 proof component

Throughout $G = \langle \rho_0, \dots, \rho_{r-1} \rangle \le A_n$, $n \ge 12$, is transitive with a connected string C-marking and every maximal omission intransitive. We assume the alternating rank bound at all strictly smaller degrees at least twelve. All supporting primitive, anchored-imprimitive, small-marking and attachment theorems below are the complete previously accepted proofs and required programs, not citations of the present source target.

Lemma 1 (A small tail and its anchor)**.** *Suppose $K = G_{>h}$ is transitive on its proper nonempty support X, with $q = |X| < 12$ and rank $d = r - 1 - h$ satisfying $d \ge (q-1)/2$. Then K is the literal five-point D_{10} of source Table 2(1), or one of its two six-point A_5 markings (2),(3), in the relevant string orientation.*

Proof. The action of K is faithful, even and has full IP, since it fixes the complement of X. Its marking is connected, being a tail of a connected string. Every maximal omission of K is intransitive on X: otherwise its corresponding global omission would already connect every point of X, and adjoining the omitted generator, supported on X, could not change any global omission orbit.

For $j < h$, the generator ρ_j centralizes K and therefore preserves its unique nontrivial orbit X. All generators except ρ_h consequently preserve X; global transitivity forces ρ_h to take some point of X outside it. The accepted extension-anchor lemma shows that this point is fixed by all tail generators except the first. Thus the small anchored-imprimitive bound excludes an imprimitive K at the stated rank. Degrees below five give no connected even transitive involutory marking: A_3 has no involutions and the involutions of A_4 generate its commuting Klein four group. The accepted small primitive attachment exclusion therefore leaves exactly Table 2 rows (1)–(3), in the relevant string orientation. □

Lemma 2 (The critical dihedral tail)**.** *Suppose a unique eligible i-edge joins omission orbits A, B, $h > i$ is the last shared label, $G_{<i}$ fixes B when $i > 0$, and $K = G_{>h}$ is the five-point D_{10} on $X \subseteq B$. If the accepted fixed-support bounds are at equality $n - |X| = 2h+1$, the configuration is impossible for $n \ge 12$.*

Proof. The permutation centralizer of D_{10} on five points is trivial: centralizing its five-cycle gives a power of that cycle, and centralizing an inverting reflection makes that power the identity. Every earlier generator commuting with K therefore fixes X pointwise.

For $i = 0$, use the exact fixed paths of the accepted support bound: A is the path $a_0, \dots, a_h$, while $b_0, \dots, b_{h-1}$ is the fixed prefix in B, followed by $b_h \in X$. Since ρ_0 fixes b_h and commutes with $\rho_h, \dots, \rho_2$, it fixes $b_1, \dots, b_h$. Its only other possible action besides $(a_0\ b_0)$ is on $\{a_1, \dots, a_h\}$. For $h \ge 3$, the subgroup generated by colours $2, \dots, h$ is transitive on that set and its subgroup of colours $3, \dots, h$ has exactly one fixed point, a_1. Centralizing both forces ρ_0 to fix the whole set, contradicting evenness. For $h = 1$ there is no nontrivial permutation of the remaining singleton, again a contradiction. Thus only $h = 2$ could remain, but then $n = 2h + 6 = 10$. This uses only the trivial centralizer of K, not its perfection.

Now take $i > 0$. In the transitive-left-side case its accepted support theorem gives $h = i + 1$, $|A| = 2h$ and fixed set $A \cup \{b\}$. The involution $\tau = \rho_h$ is the full exchange λ of h preserved pairs on A. In the intransitive-left-side case, the accepted support theorem gives the same

parameters and fixed set; A consists of two faithful natural S_h-orbits, exchanged by τ. In both cases $B = \{b\} \cup X$, and τ moves b into X.

Here is the exact perfect factor needed when h is even. Normalize the last two marks on $X = \{1, \dots, 5\}$ so the last is $l = (1\ 2)(3\ 4)$ and the preceding one is $s = (2\ 3)(4\ 5)$. Since τ commutes with l, its crossing endpoint is the unique l-fixed point 5. Its restriction to B is $(b\ 5)u$, with $u \in C_{S_4}(l)$ an involution. When h is even, λ is even, so $\tau|_B$ is even and u is odd. The only odd involutions of that centralizer are $(1\ 2)$ and $(3\ 4)$. Thus the literal projection of $R = \langle \tau, K \rangle$ to B is one of the accepted degree-six A_5 heads, and in particular is perfect. As $R \le \langle \lambda \rangle \times A_5$ projects onto both factors, perfection gives $(\lambda, 1) \in R$: otherwise R would be the graph of a nontrivial homomorphism $A_5 \to C_2$. This conclusion takes no IP in a quotient.

For the transitive-left-side case, the accepted paired-block argument gives an independent length-$h-1$ block marking of S_h for $L = G_{<i}$. If h is even, its kernel in the pair-flip space lies in the even-weight space. Every nonzero S_h-invariant subspace there contains the all-one vector: a nonconstant vector and a transposition give a weight-two vector, whose conjugates span the even-weight space; a nonzero constant vector already is all-one. But $\lambda \in R$ and actual IP gives $L \cap R = 1$. The kernel is therefore zero, so $L \cong S_h$. Its endpoint omission fixes a point and has order $(h-1)!$, whereas a transitive degree-$2h$ action of S_h has point stabilizer order $(h-1)!/2$. This is impossible. For odd $h \ge 5$, the accepted paired-row argument applies unchanged: the crossing generator fixes X, and its required odd residual involution centralizes the S_{h-1} action on the other $h-1$ pairs, forcing it even. The $h = 3$ even-vector norm argument gives an order-six group with a nontrivial point stabilizer, contradicting transitivity on six points. These are precisely the previously proved transitive-side equality arguments; their only substituted premise is the perfect factor just proved.

For the intransitive-left-side case put $k = h$. The accepted crossing-normalization lemma uses only the trivial centralizer of the tail, its support size at least five, and transitivity of R on B. It excludes odd k, and for even k gives

$$\rho_i = (u_k\ b) \prod_{j<k} (u_j\ v_j), \qquad \lambda = \prod_j (u_j\ v_j).$$

For $k \ge 4$, the accepted explicit fifth-power word and its diagonal S_k conjugates put λ in $P = G_{\le i}$. The perfect-factor argument above puts it in $R = G_{>i}$, contradicting full IP $P \cap R = 1$. The remaining $k = 2$ gives $n = 2k + 6 = 10$, outside the range. This finishes both interior cases. □

Theorem 3 (The full numerical transitive-tail alternative)**.** *Suppose the unique eligible i-edge ab joins the two G_i-orbits A, B, some label greater than i acts nontrivially on both, and h is the largest such label. For $i > 0$ assume $G_{<i}$ fixes B pointwise, as in the source's chosen interval orientation. Suppose $h < r - 1$ and $K = G_{>h}$ is transitive on its support $X \subseteq B$. Then either $r \le \lfloor (n-1)/2 \rfloor$, or an actual first or last three-generator subgroup is one of the two source Table 2 six-point A_5 markings, allowing full string reversal at the last end.*

Proof. Assume the rank bound fails. Put $q = |X|$, $d = r - 1 - h$ and $F = n - q$. The complete accepted support bounds give

$$F \ge 2h + 1. \tag{1}$$

For $i = 0$ this is the fixed-prefix bound. For $i > 0$ it is the transitive-side bound if $G_{<i}$ is transitive on A, and the completed intransitive-side bound otherwise. Here B is nontrivial because it contains X. Their exact equality conclusions will be used below.

The proper support X supplies an endpoint anchor exactly as in Lemma 1; the same proof gives connectedness, evenness and intransitivity of all tail omissions. If $q < 12$ and $d < (q-1)/2$, then integrality and (1) give $2r \le q - 2 + 2h + 2 \le n - 1$, a contradiction. For the other small case Lemma 1 leaves either an actual A_5 tail, giving the stated exception, or $q = 5, d = 2$. In

the latter case rank failure and (1) force $F = 2h + 1$; Lemma 2 excludes it. If $q \geq 12$ and K is proper primitive or imprimitive on X, the accepted primitive or anchored imprimitive theorem gives $d \leq \lfloor (q-2)/2 \rfloor$. The same inequality again contradicts rank failure.

Consequently only the full natural $K = A_X$ with $q \geq 12$ remains. The explicit smaller-degree induction gives $d \leq \lfloor (q-1)/2 \rfloor$. Together with (1) and the failure of the desired bound this forces precisely

$$q \text{ odd}, \qquad d = (q-1)/2, \qquad F = 2h+1, \qquad n = 2r. \tag{2}$$

For $i = 0$, the accepted exact alternating-tail endpoint lemma applies to the equality fixed set. It gives $h = 2$ and the actual first triple Table 2(2), with no decoration on X.

For $i > 0$ with $G_{<i}$ transitive on A, its support theorem gives $h = i + 1$, $|A| = 2h$ and fixed set $A \cup \{b\}$. The complete transitive-side alternating equality obstruction excludes this configuration.

For $i > 0$ with $G_{<i}$ intransitive on A, its completed support theorem makes the two faithful orbit sizes equal to $k = i + 1 = h$, with fixed set $A \cup \{b\}$ and diagonal S_k action on A. The accepted exact alternating-tail reduction excludes $k \geq 3$ and leaves $k = 2$: the first two marks form the pure degree-five D_{10} on $A \cup \{b\}$. The next mark exchanges the two pairs in A and exchanges b with one point of X; its remaining action is supported on X. Thus this is again an actual supported dihedral attachment, now at the first end. The complete six-choice adapter gives the rank bound or the claimed first A_5 triple. This accounts for every case under (2) and proves the theorem. □

Source and precise boundary. The theorem completes the numerical alternative of Cameron–Fernandes–Leemans–Mixer, *Highest rank of a polytope for A_n*, arXiv:1605.09173v1, Proposition 5.26, physical page 34, with its unique-crossing and chosen interval orientation explicit. Lemmas 1–2 supply the missing small dihedral-tail case by an actual anchor and a perfect-factor argument. A single exit from the five-point tail is *not* asserted to make the entire neighboring omission have two orbits; other eligible edges may lie in its complement. The source's rank-deficit strategy and its results are retained in attribution; the equality obstructions and complete dihedral adapter are the present replacement proof. All complete supporting manuscripts, finite programs and retained Whiston/classification premises remain paid. There is no new census or formal theorem, and no assumption of IP for a projected group. The intransitive-tail argument of Proposition 5.27, later source tables, connected-square rigidity and whole L044 recovery remain separate; no new original or question credit is claimed.